RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2023- 2024 (Odd Semester)

Degree, Semester & Branch	**: I Semester B.E. ECE B**
Course Code & Title	**: GE3151 Problem solving and python programming**
Name of the topic	**: Tuples**
Name of the event	**: One minute paper**

Tuple

i) declare
a = (1,2,3,4)
ii) to add elt, change to list
b = list(a)
c = b.append(5)
print c
c = (1,2,3,4,5)
iii) del a → to delete a entire tuple

Tuple

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a = (1,2,3,4)
2) to add elt, change to list
b = list(a)
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Print c
c = (1,2,3,4,5)
3) del a → to delete a entire tuple

Signature of the Faculty member

HOD

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Academic Year 2023 – 2024 (Odd Semester)
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Course Code & Title: GE3151 Problem Solving and Python Programming
Name of the Faculty member (s): P.Venkatesh

Innovative Practice Description

- **Unit / Topic:** I / Guess an integer number
- **Course Outcome:** CO1 : The students will be able to develop algorithmic solutions to simple computational problems
- **Topic Learning Outcome:** TLO 1,2,3
- **Activity Chosen:** Jigsaw
- **Justification:**
  The Jigsaw Strategy is an efficient way to learn the course material in a cooperative learning style. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. Group members must work together as a team to accomplish a common goal each person depends on all the others.
- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**
  The lessons are divided into subcategories. Students are divided into groups of four or five students. Then each small group would be created with one student receiving one Subcategory of the lesson. For this method, each small group gets the same set of Subcategories. Once individuals have researched their own subcategory, they will meet with individuals from the other small groups with the same topic to better develop their understanding and become experts of the subcategory. Each student would then return to their original group and teach their subcategory to the rest of their small group.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO8	PO9	PO10
CO1	2	3	2	3	3

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO8	PO9	PO10
	2	3	2	3	3
Justification for correlation	Guess an integer requires the knowledge of engineering fundamentals such as	The course outcome is highly correlated, since identification of	Identify tenets of the professional code of ethics while doing activity-	A Collaborative activity – Jigsaw helps the students to Present results as a	The topics deals with presentations of fixed- and floating-point representation. This task is

	representation of numbers and the order This course outcome is moderately correlated.	mathematical and engineering knowledge to solve problems related to guessing number	based learning. So, it is moderately correlated.	team, with smooth integration of contributions from all individual efforts. The course outcome is highly correlated.	accomplished through Jigsaw activity therefore course outcome is correlated highly.

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    Students feel that it was interesting. They learn how to communicate with team members and work together.
  - ***Benefit of the practice:***
    Every student got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct
  - ***Challenges faced in implementation:***
    The main challenge faced is group conflict. Few Students are not ready to work together. To avoid that conflict I started to praise others, taking turns for equal participation, and shared decision making.

References:

- **Reflective Critique:**
    - ***Feedback of practice from students and other stakeholders:***
        - Students feel that they have improved self-learning.
        - They learn how to communicate with team members and work together.
        - They were completely involved
    - ***Benefit of the practice:***
        - Every student got equal opportunity to come forward to take part in this activity.
        - They felt easy in solving level 3 questions in Assignments and Internal Assessment test
        - The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct
    - ***Challenges faced in implementation:***
        - Few students hesitated to come forward to present the results and conclusions of the disscussion
        - The main challenge faced is that few students not exposed to flipped class room.

References:

- https://www.teachthought.com/learning/a-flipped-classroom/
- Bergmann, J., & Sams, A. (2012). Flip your classroom: Reach every student in every class every day. Eugene, Or: International Society for Technology in Education.

Signature of Faculty Member **HOD**

RAMCO INSTITUTE OF TECHNOLOGY
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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

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Innovative Practice Description

- **Unit / Topic:** Unit V / break, continue, pass
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 7,8
- **Activity Chosen:** Flipped Classroom
- **Justification:**
  It allows students to learn in their own pace, it encourages students to actively engage with lecture material, it frees up actual class time for more effective, creative and active learning activities and students take control and responsibility for their learning. Also the practice was chosen because the topic is simpler and interesting for the students to refer and learn by themselves

- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**
  Specific topic was given to the students learn on their own. Resources like reference book, videos were given to the students. Apart from this they can refer any other relevant source for preparation. Students are asked to prepare more in depth than before. Students are separated into groups where students are given a specific topic to prepare and present. Get the class back together to share the individual group's work with everyone.

- **CO – PO / PSO mapping:**

CO	PO1	PO3	PO8	PO9	PO10
CO5	3	2	1	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO3	PO8	PO9	PO10
	3	2	1	1	1
Justification for correlation	Solve real time problems in topics like	PO3 is moderately mapped with level 2	Identify tenets of the professional code of	It helps the students to present the topic of	The students will present the results and conclusions of

- https://www.educationworld.com/a_curr/strategy/strategy036.shtml
https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy/

Signature of the Faculty member

HOD